Exercises Week 6

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Task 1

We have $y_t = \int K(t-t';\theta(t'))x(t')dt' + e(t)$, where $\theta(t)$ represent time-dependent kernel width. Now we want to use a discrete version for computation:

$$\begin{aligned} y_i &= K_{w,b}x_i + e_i,\ \ e_i \sim N(0,\sigma_n^2 I) \\ (K_{w,b})_{jk} &= \Delta t \frac{1}{\sqrt{2\pi\theta_k}} \exp\left(-\frac{(t_j - t_k)^2}{2\theta_k^2}\right) \end{aligned} \tag{1}$$

So that $\theta_k \approx \Psi^{NN}(t_k; w, b)$. The kth of K uses the width t' = t_k. In this task a Gaussian kernel was used.
We aim to approximate the width using a neural network with 6 hidden tanh layers, H = 6.

$$\Psi^{NN}(t; w, b) = 0.1 + \log(1 + \exp(\mathbf{W}_2 \tanh(\mathbf{W}_1 t + b_1) + b_2)) \tag{2}$$

. The value of 0.1 is a softplus offset to keep θ non-zero. In total there are 19 weights and biases, given by the equation $3H + 1 = 19$.
The weights and biases have a Gaussian priors $N(0, \sigma_p^2)$ with $\sigma_p = 1.0$. The likelihood is a product over M number of pairs

$$p(\{(y_i, x_i\}\,|w, b) = \prod_{i=1}^{M} \frac{1}{\sqrt{(2\pi)^N|\Sigma|}} \exp\left(-0.5(y_i - K_{w,b}x_i)^T\Sigma^{-1}(y_i - K_{w,b}x_i)\right) \tag{3}$$

. The posterior is prior times likelihood. Weights w and biases b are the unknown parameters. (Neal, 2012).
We can then proceed to minimize the negative log posterior with an MAP estimate

$$J(w, b) = \frac{1}{2\sigma_n^2} \sum_{i=1}^{M} ||y_i - K_{w,b}x_i||^2 + \frac{1}{2\sigma_p^2}(||w||^2 + ||b||^2) + c \tag{4}$$

, where c is some constant. The minimization of J is done with L-BFGS. I ran 5 random restarts from $\mathrm{N}(0, 0.5^2)$ and chosest the smallest J.

M	10
N	200
σ_n	0.03
l	0.1
σ_f	1

Table 1: Parameters

$RMSE(\theta, \Psi^{NN})$	10
$RMSE(Kx, K_{w,b}x)$	200

Table 2: RMSE

Exercise 2

A same model is used and prior and likelihood are set exactly like in the Task 1. The sampled posterior of $p = (w, b)$. Metropolis-Hasting algoritm is used instead of the neural network. The proposal are done with $p' = p+\text{step size}\cdot z$, where $z \sim N(0, I)$. The acceptance probability is

$$\alpha = \min(1, \exp(f(p) - f(p*)) \tag{5}$$

where is f represents the function of negative log-posterior.
To improve the mixing in the chain, I adjust the step-size half-way through the burn-in period. If the acceptance rate is below 0.20%, then the step size is reduced by 15%. If the acceptance rate is over 0.45, then the acceptance rate is increased by 15%. Essentially we try to have an acceptance rate between 0.20-0.45.(Roberts et al., 1997).

N	20000
$burn - in$	5000
max lag	100

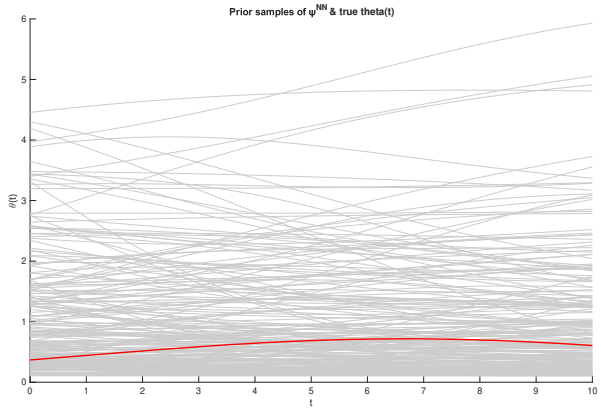
Table 3: Parameters

acceptance rate	step size	nsamples	ESS_{min}	ESS_{median}	ESS_{max}	$\text{ESS}_{min-over-N}$
0.31374	0.00535	15000	75.454	120.74	306.12	0.00503

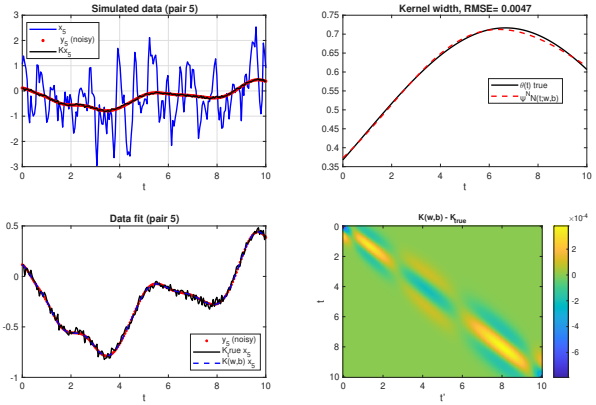
Table 4: ESS out 15 000 samples

RMSE θ_{mean} (T1)	RMSE to θ_{MAP} (T2)	Coverage
0.004686	0.004662	0.785

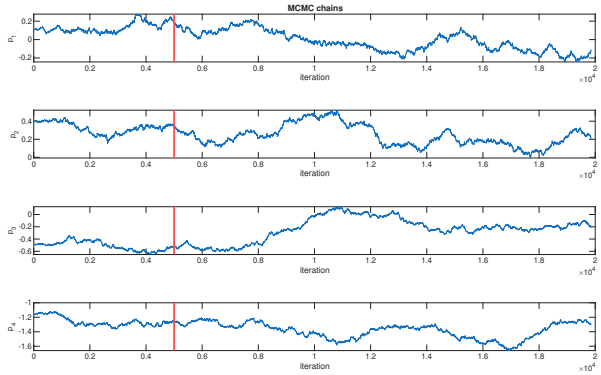
Table 5: RMSE



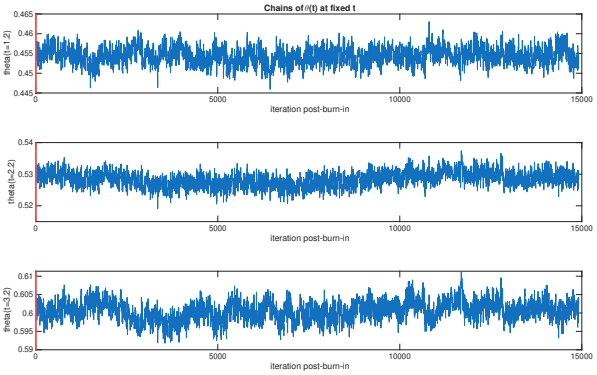
(a) Task 1: priors



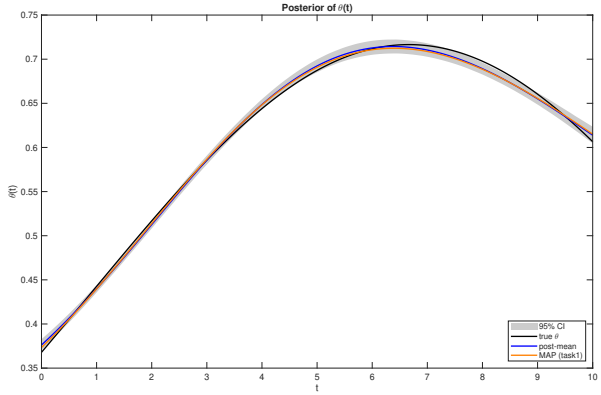
(b) Task 1:Estimates of θ



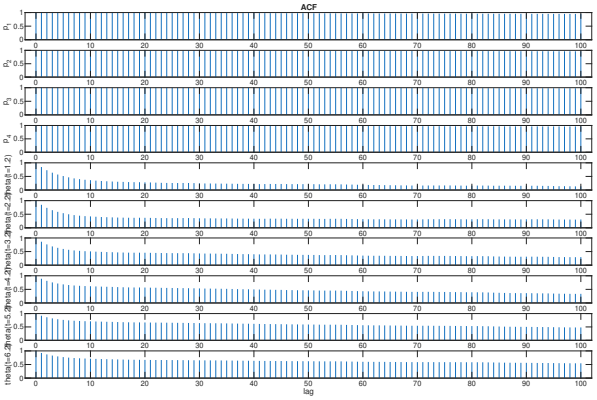
(c) Task 2: MCMC chains



(d) Task 2: Fixed θ



(e) Task 2: estimates



(f) Task 2: ACF

In Figure (a) we see the that although the samples drawn from the Gaussian prior vary greatly, they concentrate around the red curve which is the true $\theta(t)$. In Figure (b) we see that the estimates are very accurate despite there being some noise. We could add to noise level to challenge the neural network more. The diagonal in the bottom right corners show a close match between the estimator kernel and true kernel values.
In Figure (c) we have some signs of slow mixing and autocorrelation despite 5000 iterations reserved for the burn-in.
In Figure (d), there is no visible trend and the chains look like noise.
In Figure (e) are stacked on top of each other, closely following each other
In Figure (f), the decay is very slow and suggests the slow mixing as observed in Figure (c) as well. This explains the low effective sample sizes (ESS) and the coverage of only 78.5%:

References

Neal, R. M. (2012). *Bayesian learning for neural networks*. Springer Science & Business Media.

Roberts, G. O., Gelman, A., and Gilks, W. R. (1997). Weak convergence and optimal scaling of random walk metropolis algorithms. *The Annals of applied probability*, 7(1):110–120.